A Survey and Task-Based Quality Assessment of Static 2D Colormaps

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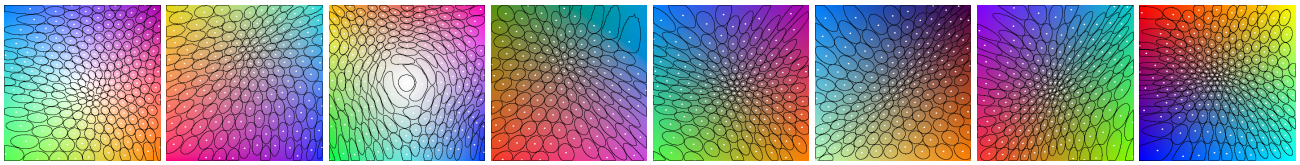


Figure 1. Static 2D colormaps with the highest overall rating for black or white backgrounds. From left to right: TeulingFig3,[1] Mittelstädt et al.,[2] Ramirez et al..[3] Bremm et al. (regular),[4] TeulingFig3NoWhitening,[1] TeulingFig2,[1] TeulingFig4a,[1] Cube Diagonal Cut B-C-Y-R,[4–8]

ABSTRACT

Color is one of the most important visual variables since it can be combined with any other visual mapping to encode information without using additional space on the display. Encoding one or two dimensions with color is widely explored and discussed in the field. Also mapping multi-dimensional data to color is applied in a vast number of applications, either to indicate similar, or to discriminate between different elements or (multi-dimensional) structures on the screen. A variety of 2D colormaps exists in literature, covering a large variance with respect to different perceptual aspects. Many of the colormaps have a different perspective on the underlying data structure as a consequence of the various analysis tasks that exist for multivariate data. Thus, a large design space for 2D colormaps exists which makes the development and use of 2D colormaps cumbersome. According to our literature research, 2D colormaps have not been subject of in-depth quality assessment. Therefore, we present a survey of static 2D colormaps as applied for information visualization and related fields. In addition, we map seven devised quality assessment measures for 2D colormaps to seven relevant tasks for multivariate data analysis. Finally, we present the quality assessment results of the 2D colormaps with respect to the seven analysis tasks, and contribute guidelines about which colormaps to select or create for each analysis task.

Keywords: 2D colormap, data comparison, data projection, color spaces, user perception, evaluation, quality assessment, information visualization, visual analytics

1. INTRODUCTION

Color is one of the most important and widely used variables to encode data values in information visualization. Prominent application fields are social sciences, statistical analysis, economics, biology, chemistry, medicine, physics, computer graphics, multimedia, and many others. Color is used to encode qualitative, as well as quantitative data values. For the latter, colormap designers distinguish between unipolar and bipolar data attributes. Influential methodologies for the use of color in general were presented by Ware and Brewer.[9,10] Further, color is not just used for univariate, but also for multivariate data. For this kind of data, dimension reduction techniques are often applied to map the data into a 2D space. As a result, static 2D colormaps can be used to facilitate the encoding of data with color, as shown in Figure 1. The analytical tasks for the use of colormaps for multi-dimensional data differ considerably compared to those for univariate data. Colormaps for univariate data are often explained explicitly with colormap legends within a visualization system. Thus, some sort of 'semantics' is introduced and color values are used to support lookup, comparison and other elementary tasks. However, the semantics (or: intrinsic properties) of multivariate data are more complex. In this case, color is usually applied in a relative context to indicate similar, or to discriminate between different multivariate data values (or clusters). Color is especially useful to

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enrich the visualization if accurate visual variables (such as position) are already used for other encodings. Moreover, in many approaches the colors of 2D colormaps are also used for linking (multivariate) data values between different views.

However, the design of 'good' 2D colormaps is a sophisticated exercise. Many quality criteria exist and requirements towards colormaps are often contradicting.[11] One example is the number of colors covered by a 2D colormap. In order to discern different data values, colormaps should contain a vast number of visually distinct colors (*color exploitation*). On the other hand, the perceived distance between two colors should correspond to the geometrical distance on the 2D colormap and thus to the distance in the data space. However, ensuring the so-called *perceptual linearity* becomes more difficult when more colors are involved. Thus, a trade-off between these two quality criteria has to be made. The challenge of designing a colormap that supports several user tasks is more difficult or may even be impossible. Following Tominski et al.,[12] the *what* aspect (what data is visualized) is not the only influencing aspect. Colormap design also depends on the question *why* (for which task[13]) the data is encoded. Furthermore, for *whom* (which users) the colormap is targeted has a decisive influence on the design. Finally, *where* the visualization is shown (device) also has considerable implications on the design.

In summary, the different criteria yield into a large design space which makes the design of 2D color maps a challenging task. While a variety of 2D colormaps exist today, we identify that these colormaps all perform differently with respect to quality assessment measures. Thus, applying 2D colormaps with respect to different analytical tasks strongly depends on the map design, and the task. One may assume that not every existing 2D colormap was designed on behalf of all important quality assessment indicators, or on the task they aim to support. However, according to our literature research, there exists no summarizing work that evaluates existing 2D colormaps with respect to quality assessment criteria.

In this work, we present a survey of the most prominent static 2D colormaps used in a variety of application fields related to information visualization. We focus on applications with general 2D colormaps where the focus is on en- and decoding data relations of more than one-dimensional data with color; and not on two-dimensional color mapping where the focus is on metric quantities of the dimensions, for example, with color-hue encoding dimension X, and saturation encoding dimension Y. The surveyed 2D colormaps were re-implemented to assess their quality. Therefore, we devise novel quality measures based on the requirements of the established task model of Andrienko & Andrienko.[13] We further present a taxonomy of task-based requirements for the use of 2D colormaps for multivariate data (note that we do not follow approaches for encoding bivariate data). For every provided task, we suggest combinations of relevant quality criteria and determine respective quality assessment measures. Finally, we present the best 2D colormaps for any of the described analytical tasks and discuss quality-dependent varieties among the 2D colormaps.

2. RELATED WORK

2.1 Colorspaces and Color Theory

How precisely color is perceived is still a focus of research. Hunt et al.[14] and Fairchild[15] offer a full discussion of color appearance. The standardized color appearance model CIECAM02[16] (which is used in this paper) is based on the results of the Hunt model but is adapted for industrial practical usage. It is well known that the RGB and HSV color spaces are not perceptually uniform, i.e. they cannot be used to measure the perceptual distance of two colors. CIE LAB and CIE LUV almost fulfil this criteria and are widely used for perceptual estimations in the field of information visualization.[17] These color spaces have been refined over the years (e.g. DIN 99) and distance metrics (such as CIEDE 2000) have been devised aiming at perceptual uniformity, where a value of $\Delta E = 1$ expresses the just noticeable difference (noticed by half of the observers). The DIN 99 color space and the CIECAM02 color appearance model are recent developments that define a perceptual color space with reversible conversion operations. In such well-defined perceptual color spaces, colormaps can be created by interpolation within the color space.

Luo et al. defined a similar ΔE based on CIECAM02,[18] based on the idea that a CAM should be a natural candidate to define a ΔE because similarity of colors is rooted in their appearance attributes. Luo et al. correlated appearance attribute differences to the color difference data sets also used to develop CIE DE 2000 and DIN99, and obtained a color difference formula and three parameterizations named CIECAM02-SCD (based on the dataset of Small Color Differences), CIECAM02-LCD (Large Color Differences), and CIECAM02-UCS (fit both datasets). In this paper, we use the CIECAM02-UCS variant (CAM02-UCS in short) as we require measures of small and large color differences which are comparable. Also, Luo et al. reported CAM02-UCS predictive performance to be on par with the SCD and LCD variants. A shortened definition is repeated in equation 1, for a complete discussion please see Luo et al..[18] CAM02-UCS is an Euclidean distance defined in a suitable derivative of CIECAM02. Starting from CIECAM02 attributes J, M and h, ΔE in CAM02-UCS is defined as follows:

$$J' = \frac{(1+100c_1)J}{1+c_1J} \quad M' = (1/c_2)ln(1+c_2M) \quad \Delta E = \Delta E_{UCS} = \sqrt{(\Delta J'/K_L)^2 + \Delta a'^2 + \Delta b'^2} \quad (1)$$
$$a' = M'cos(h) \quad b' = M'sin(h)$$

The fitting constants $K_L, c_{1,2}$ are obtained from a parametrization table determined by Luo et al. for the SCD, LCD and UCS variants. $\Delta J'$ and other deltas refer to the corresponding difference between two given colors.

2.2 Criteria for the Design of (2D) Colormaps

The existing related literature discusses many different criteria for the use of color. However, only few works have been presented discussing or evaluating design criteria for 2D colormaps. The works of Guo, and Bremm et al. may serve as examples .[4,19] Thus, in this section, we shed light on colormap design from a wide range of color-related research. In order to create a solid foundation for our measures, we review the motives, requirements and criteria introduced in the related work. The terminology differs significantly within the existing literature. Thus, we summarize different criteria with terms we consider as most representative.

Color exploitation is based on the idea that one might want to use as many colors as possible, i.e. a large (but sensible) fraction of the possible perceptible colors. Related is the phenomenon of contrast, i.e. (strongly) visible differentiation, is seen as important in several related works. Bremm et al.[4] call for 'high color resolution/contrast, [to] increase the perception of color differences for comparison'; Baum et al.[20] recommend that 'the contrast for each source should be maximized'.

Perceptual linearity: To date there is no accepted uniform color space that explains all the phenomena observable in color vision. Since we are looking at colormaps applied to data, perceptual linearity is a function of both the working color space and, for better or worse, the mapping of data values into the color space. Tominski et al.[12] elaborate 'if the data's value range is given by a linear function, the colormapping function has to be constructed in such a way that the color coding is perceived as linear as well'. Baum et al. intend linearity to express the true magnitude of variation in the data.[20]

Equal visual importance is relevant in cases where there is no prior knowledge that should be factored in, like in visualizations supporting hypothesis generation. More formally, the works of Bergmann and Rogowitz[21] suggest that for localizing and separating groups, sets of values or data points in the segmentation task, they have to be assigned with significantly different colors but with *equal visual importance*.

Separation or **separability**: Differences in values of a variable should be represented by different colors. While separability may be very dependent on qualities of the data, it often just depends on color space exploitation. Trumbo[22] stresses that a 'colormapping function f has to be invertible', which may be interpreted in the mathematical sense on f but as well along the 'whole chain', i.e. including the perceptual distance of the colors produced by f. This view gains recognition in Tominski et al.[12] who state that two different data values should be visually distinguishable because visually similar colors imply that they represent data values that are close to each other.

Rows and columns: A closely related concept is that of separation of individual color characteristics such as hue or lightness. Trumbo and Baum et al. call for an equivalent of *rows and columns* in the colormap, where value progressions of one variable should not be compromised by a second perceptual variable.[20,22] Modern color models such as CIECAM02 attempt to provide this form of separation with their color appearance attributes. While generally desirable, it is not required for mapping dimensionally reduced data, which we focus on here.

Background sensitivity is an issue when a background color is present which does not represent data, or represents a 'no data' value (known absence of data). The background needs to be visibly different from the rest of the colors used in the visualization. Thus, colormaps should avoid confusion with the background. Such criteria are generally accepted and a formal definition with application guidelines can be found in the Web Accessibility Guidelines, Section 1.4.3.[23]

Device independence, the ability to transform colors to another device or other viewing conditions, is necessary to successfully reproduce visualizations or maps. In theory this is possible if one can transform colors to CIE XYZ, the conventional color interchange space, but in practice different target devices often lead to poor color precision. Tominski et al.[12] mention this, while Brewer[10] even proof-printed maps and adjusted color schemes to achieve device indedendence.

Ease of implementation is, technically, the inverse of the complexity of the color space and the construction method. In practice it is also dependent on how understandable the accounts on how the construction is to be implemented are, or how easy they are to obtain. Many authors give only imprecise information on how they turned their data into color.

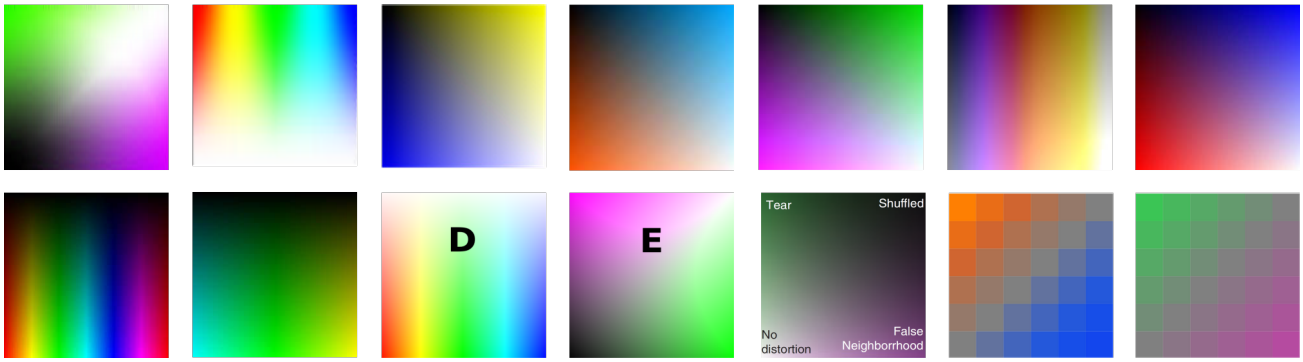


Figure 2. Colormaps excluded from the test. The designs are for the visualization of bivariate data, instead of multivariate data. From left to right, from top to bottom: Rheingans,[35] Rheingans,[35] Baum et al.,[36] Baum et al.,[36] Baum et al.,[36] Baum et al.,[37] Baum et al.,[37] Baum et al.,[37] Baum et al.,[37] Brecheisen et al.,[38] Brecheisen et al.,[38] Lespinats and Aupetit,[39] Ruppert et al. ,[40] and Ruppert et al. .[40]

2.3 Guidelines and Task Models for Colormapping

Creating colormaps for data visualizations is a well discussed topic in the literature. Ware's book[9] offers a full discussion of the visual cognition system and illustrates advanced guidelines and advice for colormapping. Silva et al. present an overview of color mapping techniques.[24,25] Colormaps for illustrative visualizations are presented by Wang et al.[26] with an overview of perceptual findings and aesthetics. These overviews and surveys primarily focus on perceptual issues and one-dimensional data encodings, which is different from our approach. Generally, there is no colormap that meets all the requirements of data visualization equally well. The challenge is to create an appropriate colormap for a specific task and specific data properties, as proposed in the literature. The resulting guidelines can be classified into data-driven and or task-driven guidelines.

A data-driven guideline has been defined by Ware.[27] He focuses on continuous data but differentiates between the metric and form tasks. For both form and metric tasks, the challenge is to encode continuous data with discrete color levels. The metric task needs an discernible representation of data values. Therefore, colormaps need to consider the just noticeable differences, that are maximized by varying over hues with linear increasing intensity.[27,28] For the form task this is an ineffective choice, since the human visual system is trained to recognize natural forms (predominantly) based on luminance.[29] This is why the gray scale colormap works very well, although it is limited to only 256 different device levels. However, contrast illusions could bias our perception when used for metric encodings.[27] An application-independent method that compensates for the bias of contrast effects in visualizations has been introduced previously by Mittelstädt et al..[30]

Bergman[21] and Rogowitz[31] present data- and task-driven guidelines. On the data side Bergman classifies type (ratio, interval), and spatial frequency (high, low). The tasks are described as isomorphic, segmentation, and highlighting. The isomorphic task faithfully reflects the structure in the (continuous) data and can be compared with Ware's metric task. For localizing and separating groups, sets of values or data points in the segmentation task, they have to be assigned significantly different colors but with equal visual importance. Colormaps for segmentation and categorical data have been discussed previously.[17,32]

Tominski et al.[12] based their guidelines on a task model of Andrienko and Andrienko.[13] The visualization tasks are grouped in two levels: *elementary* and *synoptic* tasks. While elementary tasks address individual data points, synoptic tasks consider sets of values or data points. On a lower level, elementary and synoptic tasks are further classified into lookup (identification or localization of data values or sets of values), comparison of absolute and relative values or between sets of values, and relation seeking (search for occurrences of relations specific between data characteristics).

Multi-dimensional encodings with color were evaluated by Ware[33] and Wainer.[34] Wainer evaluated two-dimensional encodings and found that 2D colormaps are unintelligible to read metrics of both dimensions. Contradicting, Ware evaluated whether color dimensions support the user in cluster identification in comparison to spatial dimensions and found that each additional color dimension is as useful as each additional spatial dimension. Mittelstädt et al.[11] therefore grouped the tasks of Tominski et al.[12] and supported cluster identification and cluster localization with their colormapping optimization.

The works surveyed for this paper were distributed in a variety of application domains following several different analysis tasks. Following the definitions above, the most prominent task observed was the *comparison* task on the granularity of both single data objects (elementary tasks), and clusters (synoptic tasks).[13]

2.4 List of 2D Colormaps

Table 1 shows a list of 2D colormaps presented in the related work. A total of 22 colormaps are listed. For every colormap, the title, a short description, and the color space is shown. In the column 'REAL' an overview of all colormaps is given. The static 2D color distribution of every colormap also forms the basis for the subsequently conducted quality assessment tests. Table 1 also displays the (local) value distribution of features of every 2D colormap. The features are in order of appearance: luminance, red channel, green channel, blue channel, hue values, saturation, value, and attention steering. We neglected the class of 2D colormaps which target the visualization of two-variable relations as frequently applied in the geo-visualization or simulation context. In Figure 2, we present representatives of this group.[35–42]

3. APPROACH

In this section, we detail our 2D colormap quality assessment strategy. Starting with a description of the most relevant analytical tasks for the use of 2D colormaps for multivariate data, we define a set of quality assessment measures which can be applied to static 2D colormaps. We then demonstrate how to map each of the measures to the analytical tasks and vice versa. As a result, we obtain a test bench to evaluate colormap applicability with respect to a set of analytical tasks.

3.1 Requirements for Applying 2D Colormaps based on a Set of Analytical Tasks

The general goal of visual analysis tools is to provide means to perform certain analysis tasks. Thus, colormaps should be selected according to requirements that support these tasks. We exclude the discussion of the data properties in this paper and focus on the general analysis tasks and discuss how they can be effectively supported by colormaps in general. The scope of the paper is on multi-dimensional visual analysis where the focus is on perceiving the data relations in the visual mapping. Our requirements are based on the task model of Tominski et al.[12] In the following, we describe each analysis task in detail and define the corresponding requirements. In accordance with Tominski's work, we group the tasks in two levels. *Elementary tasks* are defined on single objects in the visualization with respect to their visual mapping whereas *synoptic tasks* are performed on sets of objects (clusters or classes). Requirements for colormaps for the *elementary tasks* are as follows:

ER1 **Elementary Lookup** comprises localization and identification. Elementary localization defines the task of detecting single objects in the visualization with respect to their properties. For example, finding objects that are encoded with a light blue color. Elementary identification defines the task of identifying the properties of single objects within the visualization. An ideal colormap therefore provides n distinct and salient colors for n distinct objects that are visually separable. The requirement **ER1** is therefore that colormaps have to provide as many colors as possible that are distinct with respect to human perception.

ER2 **Equal Visual Importance** is the requirement that colors are equally salient. In elementary lookup tasks, especially in exploratory setups, analysts may be biased if single objects stand out due to perceptual color effects not related to object properties. This requirement is not valid for highlighting tasks, where certain objects should be highlighted with e.g., a special color. Therefore, **ER2** defines that colormaps should exhibit equal visual importance for every color.

ER3 **Elementary Comparison** defines the task of comparing the properties of two or more objects. Here, the analyst compares the colors of objects and decides whether they are similar or different. Elementary relation seeking describes the task of localizing data relations between objects in the visualization. The requirement here is that the data relations are preserved in the visual mappings. In other words, the distance of the objects in data space should be equivalent to the perceived color distance in the visualization space. Therefore, a conforming 2D colormap has to be perceptually linear.

Synoptic tasks are focused on the analysis of sets or groups of objects e.g., identifying clusters in the visualization according to their color-encoded properties. However, the visualization of groups still represents the sets of single data elements. Thus, requirements that support the effective mapping of properties of single objects also serve as a basis for higher level tasks.

SR1 **Synoptic Localization** defines the task to detect an (a priori known) group or class in the visualization. This implies that one group has to be visually differentiable against all other groups. For example, one class is encoded in blue whereas the other class is encoded in red. The color mapping has to provide colors that separate (known) groups or classes of data objects by color (**ER1**) and still satisfy **ER3** between sets. As discussed for elementary localization, attention steering towards certain elements or groups can also be harmful and has to be avoided (**ER2**).

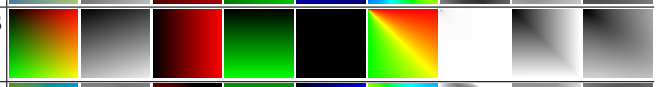
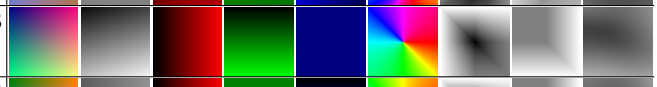
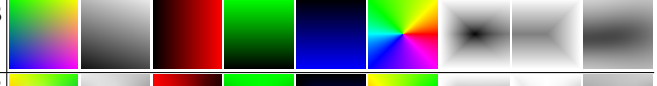
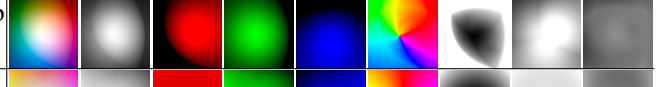
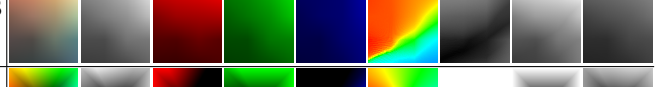
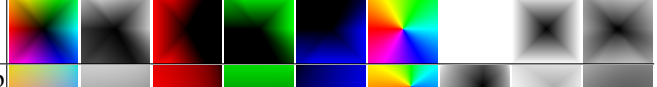
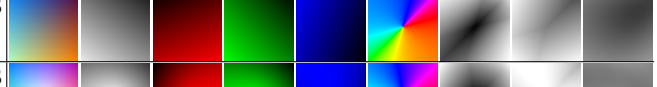
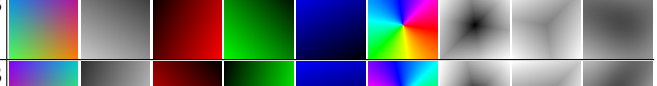
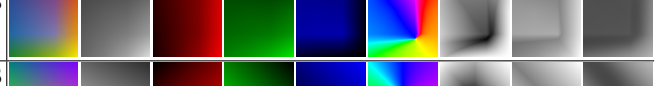
Title	Space	REAL	LUM	R	G	B	HUE	SAT	VAL	ATT
BCP 37[43] CIELab colormap with colors from the BCP 37 (Berkeley Color Project). Implementation based on a sRGB regression model	sRGB									
Baum et al. G-Y-R-B[20,37] 2D color lookup table by Baum et al. with the 4 color anchors Green, Yellow, Red, Black	sRGB									
Bremm et al. (regular)[4] Bremm et al. 2011 Fig. 9a: assumed numbers, stick to description except for a shift.	L*a*b									
Bremm et al. (stretched)[4] Bremm et al. 2011 Fig. 9b: assumed numbers.	L*a*b									
Constant Blue[4,44–47] RGB colormap with constant Blue. Red and Green span one axis each.	sRGB									
Constant Green[4] RGB colormap with constant Green. Red and Blue span one axis each.	sRGB									
Constant Red[4] RGB colormap with constant Red. Green and Blue span one axis each.	sRGB									
Cube Diagonal Cut B-C-Y-R[4–8] RGB cube diagonal cuting with the anchors Yellow, Red, Blue and Cyan. X-axis: Green, y-axis: Blue vs. Red	sRGB									
Cube Diagonal Cut B-M-Y-G[4,48–51] RGB cube diagonal cuting with the anchors Green, Yellow, Magenta, and Blue. X-axis: Red, y-axis: Blue vs. Green	sRGB									
Four Corners R-B-G-Y[52–54] RGB colormap with four color anchors in the edges: Red, Blue, Green, and Yellow.	sRGB									
Griscom and Palmer[43] Griscom and Palmer 2013 - not really a 2D color map, maybe not Lab.	L*a*b									
Guo et al. (Gaussian)[55,56] Guo 2005 (L*a*b*; Gaussian lightness; xy mapped to ab)	L*a*b									
Ramirez et al.[3] HSL-based map with white center and maximal saturation.	HSV									
Robertson and OCallaghan[57] RGB colormap with 4x4 discrete color anchors. In the corners: Light-Orange, Brown, Dark-Brown, Blue	sRGB									
Simula and Alhoniemi[58] HSL-based map with black center and constant saturation.	HSV									
Mittelstädt et al.[2] Generic CIELab-based colormap with perceptually (almost) equi-distant corners	L*a*b									
TeulingFig2[1] See Teuling paper fig. 2 or 4(b). Orientation as in Fig. 4.	sRGB									
TeulingFig3[1] See Teuling paper fig. 3 / 4 (c). This colormap is as described, not as shown in the paper. Orientation as in Fig. 4	sRGB									
TeulingFig3NoWhitening[1] See Teuling paper fig. 3 / 4 (c) upper part.	sRGB									
TeulingFig4a[1] See Teuling paper fig. 4(a). This is as described geometrically, which is not really in line with the shown colormap.	sRGB									
Wainer and Francolini[34] RGB colormap with 4x4 discrete color anchors. The colors in the corners are: Red, Blue, Green, and Yellow.	sRGB									
Yeo et al.[59] Topological map in geometric RGB color space. Colors: Green, Dark Green, Pink, Purple	sRGB									

Table 1. An overview of all re-implemented colormaps. The images show the colormap as described in the corresponding paper, and the spatial distributions of different color components: the luminance, red, green and blue, the hue, saturation, value, and attention steering.

SR2 In **Synoptic Identification**, the task is to identify (unknown) groups or classes in the visualization. For example, an analyst identifies clusters of different reddish colors sharing similar properties. Therefore, the analysts' attention should not be dragged towards distracting visual encodings of certain groups. Therefore the visual mapping has to preserve all pair-wise data relations (**ER3**) and further does not assign some groups a high visual importance (**ER2**).

SR3 **Synoptic Comparison** and relation seeking are tasks to compare two or more sets of objects. This implies that the properties within a set or cluster is preserved and also the relations between clusters. This transfers **ER3** to two different levels. The color mapping should preserve global (between sets) and local (within sets) data properties. If global relations are in focus, the requirement of preserving *all* pair-wise distances (**ER3**) can be relaxed since local data-properties of single elements are not compared across sets. If the task is to relate or to compare sets of objects, then **SR3** requires that the relations between sets have to be preserved in the coloring.

SR4 Corresponding to **SR3**, if the focus is on analyzing local properties of sets, then local relations between set elements have to be preserved. Therefore, the coloring can also relax **ER3** of preserving *all* pair-wise distances and focus on preserving local relations and discarding global (between sets) relations. Thus, **SR4** can be defined to preserve the relations of an object to its neighborhood in the color mapping.

3.2 Quality Assessment Measures for 2D Colormaps

The related work on color theory and colormap design has presented a variety of different quality criteria. We provide a set of quality assessment measures which are mapped to the requirements for applying 2D colormaps. As can be seen in the related work (e.g., Bremm,[4] Mittelstädt[11]), one of the most critical issues in the design of colormaps concerns the trade-off between color space exploitation and perceptual linearity. All measures are data-independent but focus on the static 2D space exploited by of the 2D colormaps. The measures calculate scalar values for every 2D colormap. As a result, these measures facilitate the comparability of 2D colormaps at a global scale, all shown in Table 2. A possible drawback is that quality criteria hidden in local structures of the 2D colormaps may be overlooked.

Color Space Exploitation - JND Colors (Just Noticeably Different Colors) Color space exploitation is relevant for elementary **ER1** and synoptic **SR1** lookup tasks, which includes the localization and identification of objects or sets of objects. The ability of encoding dissimilar objects with significantly different colors is crucial for a good 2D colormap. Thus, 2D colormaps should exploit large parts of the available color space confirming the colormaps a large number of distinct colors. The just noticeable distance of two colors is defined in color spaces such as CIE LAB (cf. Section 2.1). Thus, two colors are perceptually different (distinct) if the color difference ΔE is greater than a predefined threshold depending on the color space. According to Sharma et al.[60] a value of 2.3 is roughly equal to the JND in CIELAB. Therefore, we apply $\Delta E > 3$ as a parameterization for the JND colors to estimate the number of perceptually distinct colors for obtaining a measure for the exploitation of available colors. We define a *JND region* on a colormap as the spatial area where the perceived distances to the center of the region satisfy the equation $\Delta E <= \frac{3}{2}$ for all contained color points. This allows us to define a density-based measure for a JND region membership similar to compactness measures for cluster quality assessment. We calculate the number of JND colors by estimating the number of non-intersecting JND regions within a 2D colormap beginning at the center of a colormap (cf. Figure 1). The relative context of applying the identical algorithm to every colormap confers the result a sufficient degree of robustness. We are aware that other implementation variants may lead to slightly different results. However, the class of algorithms estimating the number of JND colors provides a powerful means to assess the color space exploitation.

Distance to Background - Black Distance and White Distance Most visualization systems draw their color-encoded data onto a background. Just as well, most papers are printed on white paper. Some colors are more or less visually salient on different backgrounds (bright colors are well visible on dark backgrounds but their visibility is reduced on white background). Therefore, the background reduces the amount of colors that can carry information to the analyst. Likewise, we identify the problem of colors which are similar to contours colors, text, or labeling colors. In such a case, shapes cannot be differentiated from colored objects preattentively. We provide two quality assessment measures which calculate the minimum JND distance of every colormap to a white and black reference color *rc*.

$$BC(cm, rc) = \min_{c \in cm} \left(\Delta E\,(c, rc) \right) \tag{2}$$

JND Region Size Local perceptual linearity is expected for **SR4** where the focus is on analyzing the relations of elements within sets. We assume that all sets within the data span over the whole colormap, which implies that the elements of one set will span over a smaller region of the colormap. Therefore, one can relax **ER3** of preserving *all* pair-wise distances and focus on preserving local relations, which implies that only local areas of the colormap need to provide perceptual linearity. The size of JND regions indicates the homogeneity of the color distribution in the map. If a colormap would be perfectly perceptually linear, all JND regions would have the same size since the ratio of all pairwise perceptual distances and respective spatial distances is equal. We define the *Region Size* measure as the standard deviation of all JND region sizes of a particular 2D colormap. We invert the scores since lower standard deviations indicate higher quality.

Visual Importance - Attention Steering Not all elements in a visualization are of equal visual importance. Our attention is dragged towards objects that stimulate our senses. As an example, in social life *signal colors* are used to gain human attention. To satisfy **ER2** and **SR2**, we measure whether there are differences in the visual importance of colors within a colormap. Many experiments have been conducted to determine which color properties drag the human attention. As a result, color temperature and certain hues,[26,61] as well as brightness and saturation[62] attract attention. This also correlates to the guidelines and rule-of-thumbs of the information visualization domain (e.g.,[6326]). Yet, no measure was defined to validate the potential attention of a color. However, since the experiments of Brewer[63] and Camgös[62] reveal the strong impact of brightness and saturation, we adapt the method of Guo[19] that uses a bell-shape raster in CIELAB to create diverging colormaps. This results in equivalent lightness and chroma for all colors and potentially assigns each color the same importance. Therefore, we define *attention ca* of a color c with the hypotenuse between lightness L and chroma C in CIELAB with $c = \sqrt{a^2 + b^2}$. To measure whether there are differences in the visual importance of colors, we compute the standard deviation of the visual attention of a set of sampling points. High standard deviations constitute high violations of **ER2** which is why we invert the score.

$$ca(c) = \sqrt{L_c^2 + C_c^2} \tag{3}$$

$$ca_{cm} = \frac{1}{|cm|} \sum_{c \in cm} |ca(c) - \bar{ca}| \tag{4}$$

Distance Ratio Divergence - Perceptual Linearity Providing perceptual linearity is the most relevant requirement for the comparison of data points, relevant for **ER3 SR3**. The arithmetical distance of two (multivariate) objects (e.g., provided by distance measures) should be proportional to the perceived color distance. In other words, similar objects in data space should have similar colors, while dissimilar objects should have dissimilar colors. For the use of static 2D colormaps, multivariate data are typically projected into 2D. Thus, for the quality of static 2D colormaps it is crucial that the spatial distances of any two points on a 2D colormap are proportional to the perceived color distance. If the perceptual distance between the data objects is equivalent to the spatial distance, then the colormap has the property of perceptual linearity. This measure also tests for the preservation of global (between sets) relations, since we assume that all sets within the data span over the whole colormap and thus, tests for **SR3**. We can define perceptual linearity based on the ratio ph between the color distance as measured in ΔE and the arithmetical distance (see above) for a pair of projected points p_1, p_2 and their corresponding pair of colors c_1, c_2 in the colormap cm.

$$ph_{cm} = \frac{\Delta E(c_1, c_2)}{|p_1 - p_2|} \tag{5}$$

One measure to assess the perceptual linearity is the *Distance Ratio Divergence*. In a perfect colormap, the ratio ph is equal for every possible pair of points. We apply Monte Carlo sampling to provide arbitrarily distant pairs of points and sample the pm_{cm} for each pair. The Distance Ratio Divergence measure is then defined as the standard deviation of the ph variable.

$$\sigma_{cm}^2 = \sum_{i=1}^{N} \Big(ph_{cm}(MC_i) - \emptyset\big(ph_{cm}(MC_i)\big) \Big)^2 \tag{6}$$

Distance Ratio Quantiles - Perceptual Linearity We assess the perceptual linearity from a complementary perspective. We focus on the so-called 'lie factor' which is the factor of mis-judgment that a visualization may evoke. In contrary to the *Distance Ratio Divergence* which considers all ph_{cm}, we here concentrate in *outlier values* to assess the most diverging perceptual linearity effects in a colormap. From an identical Monte Carlo sampling, we again use ph_{cm}. For *Distance Ratio*

Quantiles, we take the 5% and 95% quantiles of all ph_{cm} and provide their quotient as a measure of divergence. In effect, this measure states the worst lie factor to assume in 90% of possible color distance-based judgements. Good values are just above 1. On the contrary, values greater than 5 reflect problems in the colormap design method or parametrization.

$$DRD = \frac{Q_{0.95}}{Q_{0.05}} \tag{7}$$

4. GUIDELINE FOR THE TASK-BASED CHOICE OF STATIC 2D COLORMAPS

In the following, we discuss the results of our quality assessment measures applied on 22 re-implemented 2D colormaps. A matrix showing the results for all seven quality assessment measures is shown in Table 2. The overall goal was the creation of a repeatable guideline for the task-based choice of static 2D colormaps that is being backed by reproducible measures.

In Section 3, we characterized requirements that support most relevant analysis tasks. We suggest a mapping of the seven presented quality measures to the analysis tasks, depending on the ease of use for each of the provided tasks. A matrix with all 22 colormaps showing a guideline for the analysis tasks is shown in Table 3. Since every task has a relevant subset of the seven quality measures, we apply the rank-based Borda count election method to obtain an overall ranking of colormaps for each analysis task. The rank-based means of the election method is robust in comparison to normalization-based methods. The Borda count criterion assigns points to each candidate and quality measure, until the count of each colormap defines the overall rank. To increase the robustness of the method, we filter out colormaps with worse performance in single quality measures. In particular, colormaps performing worse than 80% other candidates in a respective quality measure are assigned an overall score of $\notin$. Thus, the applied method is comparably robust, prefers good overall performance, and punishes quality flaws in single quality measures. However, other election criteria are possible. We divide our task-based guideline for the usage of black and white backgrounds. As a result, colormaps do not necessarily have to perform well for both black and white backgrounds. As an example *Simula and Alhoniemi* exploits large parts of the black color space resulting in high numbers of JND colors but a weak Black Distance score. However, the colormap is appropriate for white backgrounds.

4.1 Elementary Localization and Identification (ER1)

Relevant Measures: JND Colors, Black / White Distance

Users perform elementary localization and identification tasks, whether they try to locate items based on a given color or identify the color of a data point. As a consequence a high number of JND colors is deemed important. Moreover, the background and contour colors of the visualization play a major role, which are covered by the Black and White Distance category. On black backgrounds *Cube Diagonal Cut B-C-Y-R*, *Ramirez et al.*, and the *Teuling et al.* (beside *TeulingFig2*) perform well. We recommend to use *TeulingFig3* for this task on black backgrounds. On white backgrounds *Baum et al.*, *Cube Diagonal Cut B-C-Y-R*, *Simula and Alhoniemi*, *TeulingFig3NoWhitening*, and *TeulingFig4a* perform well, whereas *Simula and Alhoniemi* has the best performance. Especially *Cube Diagonal Cut B-C-Y-R* shows good performance on both black and white backgrounds, and is easy to re-implement.

4.2 Equal Visual Importance (ER2)

Relevant Measures: Attention Steering

The requirement for this task is to provide constant visual importance across the entire map to prevent attention steering effects in elementary lookup tasks. Maps that perform best in the Attention Steering category are *BCP 37*, the L*a*b* maps of *Bremm et al.*, *Griscom and Palmer*, and *TeulingFig3*. However, beside *Bremm et al. (regular)* and *TeulingFig3* these maps have low scores in the color exploitation and do not satisfy **ER1**. To conclude, we recommend *Bremm et al. (regular)* on bright backgrounds and *TeulingFig3* on dark backgrounds in the elementary lookup tasks when the avoidance of attention steering effects is relevant to the application.

4.3 Elementary Comparison (ER3)

Relevant Measures: Black / White Distance, JND Region Size, Distance Ratio Divergence

For elementary comparison, perceptual linearity (global as well as local) is very relevant. Most colormaps that are created in perceptual uniform spaces such as CIELAB are suited for this tasks since color distance equals spatial distance in the colormap and thus, encolored data points can be perceptually compared. Interestingly, *BCP 37* has good scores although the colormap is created in sRGB, which is due to the small coverage of the color space. Since *Bremm et al. (regular)* and *Mittelstädt et al.* share high scores for perceptual linearity, we recommend *Bremm et al. (regular)* for elementary comparison on bright backgrounds and *Mittelstädt et al.* on dark backgrounds.

Title	Space	Img.	JND Colors The number of colors with a pair-wise distance of $\Delta E >= 3.0$	BlackDistance The smallest found JND distance to black	WhiteDistance The smallest found JND distance to white	Region Size The standard deviation of the JND region size	Attention Steering The standard deviation of the attention steering (Lightness and Chroma) variable	Dist. Ratio Divergence The standard deviation of the color distance/map distance ratio	Distance Ratio Quantiles Ratios of color distances to spatial distances. Quotient of 0.95 and 0.05 quant. to assess the divergence of linearity
BCP 37	sRGB		136	53.2	39.6	-15.5	-5.5	-12.0	-2.5
Baum et al. G-Y-R-B	sRGB		437	5.9	53.9	-11.0	-17.2	-23.2	-2.7
Bremm et al. (regular)	L*a*b		368	54.7	45.2	-8.6	-6.2	-14.9	-2.0
Bremm et al. (stretched)	L*a*b		200	54.7	45.2	-26.4	-3.0	-14.6	-3.2
Constant Blue	sRGB		435	39.2	35.5	-10.1	-11.1	-23.3	-2.9
Constant Green	sRGB		359	52.0	39.1	-13.5	-7.2	-18.0	-2.5
Constant Red	sRGB		492	41.1	35.6	-9.2	-12.8	-25.4	-2.7
Cube Diagonal Cut B-C-Y-R	sRGB		585	47.8	41.8	-7.1	-11.5	-23.5	-2.4
Cube Diagonal Cut B-M-Y-G	sRGB		470	52.4	42.7	-11.6	-14.7	-35.8	-4.1
Four Corners R-B-G-Y	sRGB		584	45.8	24.5	-16.3	-13.2	-46.7	-5.7
Griscom and Palmer	L*a*b		193	54.7	45.2	-16.1	-2.3	-12.8	-2.3
Guo et al. (Gaussian)	L*a*b		998	31.5	5.8	-5.2	-6.4	-33.2	-2.7
Ramirez et al.	HSV		458	60.8	11.5	-6.7	-6.3	-21.3	-2.5
Robertson and OCallaghan	sRGB		190	25.2	27.6	-31.0	-9.5	-13.9	-3.0
Simula and Alhoniemi	HSV		1027	8.2	41.8	-4.5	-16.7	-39.9	-3.1
Mittelstädt et al.	L*a*b		423	60.3	29.2	-4.4	-6.5	-16.1	-1.9
TeulingFig2	sRGB		446	33.4	36.5	-6.6	-9.0	-15.7	-2.0
TeulingFig3	sRGB		547	61.4	5.9	-5.4	-5.9	-17.0	-2.0
TeulingFig3NoWhitening	sRGB		442	52.9	47.4	-7.5	-9.1	-19.1	-2.3
TeulingFig4a	sRGB		459	53.0	45.7	-7.9	-11.1	-23.4	-2.6
Wainer and Francolini	sRGB		365	43.9	43.9	-20.9	-8.9	-46.2	-6.9
Yeo et al.	sRGB		370	52.5	42.5	-10.4	-12.0	-28.7	-3.2

Table 2. Summary of all quality assessment measures. A continuous color scale from green to gray indicates the quality from good to bad.

Title	Color Space	Image	Black Background							White Background							
			ER1 Elementary Lookup	ER2 Equal visual importance	ER3 Elementary Comparison	SR1 Synoptic Localization	SR2 Synoptic Identification	SR3 Synoptic Comp. (local)	SR4 Synoptic Comp. (global)	ER1 Elementary Lookup	ER2 Equal visual importance	ER3 Elementary Comparison	SR1 Synoptic Localization	SR2 Synoptic Identification	SR3 Synoptic Comp. (local)	SR4 Synoptic Comp. (global)	Min (SR1, SR2) MotionExplorer[53] (UseCase)
BCP 37	sRGB		∉	++	+	∉	∉	∉	∉	∉	++	+	∉	∉	∉	∉	∉
Baum et al.	sRGB		∉	∉	∉	∉	∉	∉	∉	++	∉	∉	∉	∉	∉	∉	∉
Bremm et al. (regular)	L*a*b		+	++	++	++	++	++	Ø	+	++	++	++	++	++	Ø	++
Bremm et al. (stretched)	L*a*b		∉	++	∉	∉	∉	∉	∉	∉	++	∉	∉	∉	∉	∉	∉
Constant Blue	sRGB		−	−	−	−	−	Ø	Ø	−	−	−	−	−	−	Ø	−
Constant Green	sRGB		∉	+	Ø	∉	∉	∉	∉	∉	+	−	∉	∉	∉	∉	∉
Constant Red	sRGB		Ø	∉	−	∉	∉	∉	∉	+	∉	−	∉	∉	∉	∉	∉
Cube Diagonal Cut BCYR	sRGB		++	−	Ø	+	Ø	Ø	+	++	−	Ø	++	Ø	−	+	Ø
Cube Diagonal Cut BMYG	sRGB		+	∉	∉	∉	∉	∉	∉	+	∉	∉	∉	∉	∉	∉	∉
Four Corners R-B-G-Y	sRGB		+	∉	∉	∉	∉	∉	∉	∉	∉	∉	∉	∉	∉	∉	∉
Griscom and Palmer	L*a*b		∉	++	∉	∉	∉	∉	∉	∉	++	∉	∉	∉	∉	∉	∉
Guo et al. (Gaussian)	L*a*b		∉	+	∉	∉	∉	∉	∉	∉	+	∉	∉	∉	∉	∉	∉
Ramirez et al.	HSV		++	+	++	++	++	+	++	∉	+	∉	∉	∉	∉	∉	∉
Robertson and OCallaghan	sRGB		∉	Ø	∉	∉	∉	∉	∉	∉	Ø	∉	∉	∉	∉	∉	∉
Simula and Alhoniemi	HSV		∉	∉	∉	∉	∉	∉	∉	++	∉	∉	∉	∉	∉	∉	∉
Mittelstädt et al.	L*a*b		++	+	++	++	++	+	++	−	+	++	−	++	+	++	−
TeulingFig2	sRGB		∉	Ø	∉	∉	∉	∉	∉	Ø	Ø	++	+	++	+	++	+
TeulingFig3	sRGB		++	++	++	++	++	++	++	∉	++	∉	∉	∉	∉	∉	∉
TeulingFig3NoWhitening	sRGB		+	Ø	+	+	+	Ø	+	++	Ø	++	++	++	+	+	++
TeulingFig4a	sRGB		++	−	Ø	++	Ø	Ø	+	++	−	+	++	+	Ø	+	+
Wainer and Francolini	sRGB		−	Ø	∉	∉	∉	∉	∉	Ø	Ø	∉	∉	∉	∉	∉	Ø
Yeo et al.	sRGB		−	−	Ø	−	−	−	Ø	Ø	−	−	−	−	−	Ø	−

Table 3. Applicability of static 2D colormaps for analytical tasks, for black and white background colors. We suggest combinations of quality measures (cf. Tab. 2) for every task. The Borda count election method provides the overall scores aggregated to ++, +, Ø, and − in the order of applicability. We exclude colormaps if outnumbered by more than 80% of the competitors in one relevant metric ∉.

4.4 Synoptic Localization (SR1)

Relevant Measures: JND Colors, Black / White Distance, Attention Steering, JND Region Size, Distance Ratio Divergence.

With the synoptic localization task users are expected to detect an (a priori known) group or class in the visualization. This requires that the relations of data points are preserved in the colormapping, which in turn requires preserving perceptual linearity. However, the most relevant qualitative requirement for this task remains the number of distinct colors in order to differentiate between groups to ease localization. Therefore, six quality assessment measures were applied in the test setup, whereas the weighting of the *JND Colors*, and the *Background Distances* measures were emphasized (the influence of these measures was doubled in the sum of all involved measures). The 2D colormaps with the best quality assessment scores are *TeulingFig3* (black backgrounds) and *TeulingFig3NoWhitening* (white backgrounds). Both colormaps provide an excellent trade-off between color space exploitation and perceptual linearity with an emphasis on the number of distinct colors.

4.5 Synoptic Identification (SR2)

Relevant Measures: JND Colors, Black / White Distance, Attention Steering, JND Region Size, Distance Ratio Divergence.

Identifying groups of data points in a visualization by interpreting colored information is a very complex task. The coloring has to preserve data distances of data points in order to faithfully represent data relations and not to produce artifacts such as similar colors for dissimilar data points. At the same time, the coloring should exploit the color space in order to allow discriminating between groups. Similar to synoptic localization, this requires a trade-off between perceptual linearity and color space exploitation. In this task however, perceptual linearity is the major requirement, since the groups are unknown and faithfully encoded data relations help to reveal the groups in the coloring. Therefore, six quality assessment measures were applied in the test setup, whereas the weighting of the *Region Size*, and the *Distance Ratio Divergence* measures were emphasized (the influence of these measures was doubled in the sum of all involved measures). In the final ranking, *TeulingFig3* (dark background) and *Mittelstädt et al.* (dark and bright background) share the highest rank in this task. Interestingly, *Mittelstädt et al.* has the highest rank on white backgrounds although it does generally not perform well on this background. This indicates a clear need for colormaps that support synoptic identification on white backgrounds.

4.6 Synoptic Comparison and Synoptic Relation (SR3–global) (SR4–local)

Relevant Measures: JND Colors, Black / White Distance, Attention Steering, JND Region Size (local), Distance Ratio Divergence (global).

As mentioned above, the tasks of synoptic comparison and relation seeking can be transferred to two different levels: between sets (global, e.g., comparing clusters) and within sets (local, e.g., comparing cluster elements). Therefore, different measures of perceptual linearity should be applied to accurately estimate the support of the colormaps for those two levels. As with other synoptic tasks, the trade-off between perceptual linearity and color space exploitation is obligatory, whereas perceptual linearity plays a major role and is emphasized for these two tasks, accordingly. While *Region Size* estimates the variance of perceptual linearity locally, it is used to estimate the performance for applications that require comparison within sets. *Distance Ratio Divergence* is a measure for global perceptual linearity and thus reveals the support of between–set comparison. *Bremm et al. (regular)* has the highest rank for global tasks and *Mittelstädt et al.* scores best for local tasks.

5. USE CASE

We proceed to show the applicability of our guideline for the task-based choice of static 2D colormaps. The MotionExplorer[53] system serves as a use case example. MotionExplorer supports domain experts in human motion analysis by providing exploratory search capability for large data collections. Based on two overview visualizations (not shown) the search for motion sequences begins with querying a start and an end cluster by-example. The retrieved sequences (here: jumping jack motions) are shown in the Figures 3, 4, and 5. At the center, retrieved motions are shown, visually encoded with colored sequence bars. A grid-based color legend (left) provides a mapping of clustered human poses (the high-dimensional feature vectors) to colors of a static 2D colormap. Thus, colors allow for synoptic localization (**SR1**) and identification (**SR2**) tasks in different linked views. The best colormap in the use case is expected to provide the highest minimum score in the tasks **SR1** and **SR2** for a white background (cf. Table 3). For the sake of completeness, the underlying quality measures for **SR1** and **SR2** are JND Colors, White Distance, Region Size, Attention Steering, and Distance Ratio Divergence (cf. Table 2). 14 of the colormaps are excluded for **SR1** or **SR2**, just like for MotionExplorer. One of the inappropriate colormaps for the given tasks is e.g., *Robertson and OCallaghan* shown in Figure 3. The two colormaps *Bremm et al. (regular)* and *TeulingFig3NoWhitening* shown in the Figures 4 and 5 score with ++ .

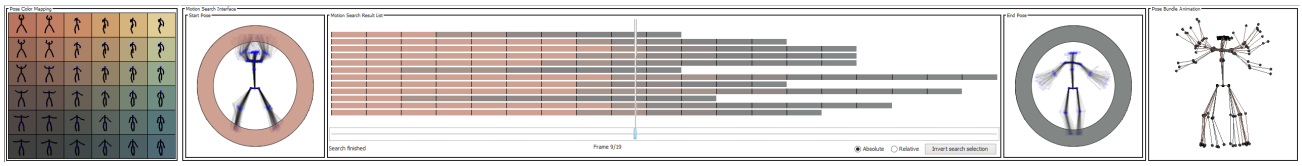


Figure 3. An inappropriate colormap for the tasks performed in MotionExplorer: Robertson and OCallaghan.

The 190 JND Colors of *Robertson and OCallaghan* are low compared with *Bremm et al. (regular)* and *Teuling-Fig3NoWhitening* (368 and 442 JND Colors). The White Distance of *Robertson and OCallaghan* is comparably weak, see the upper right corner of the colormap. In contrary all regions of *Bremm et al. (regular)* and *TeulingFig3NoWhitening* provide clear differences to white colors, in Table 2 these differences are measured with at least 45.2 and 47.4 JND Colors. The scores for the Region Size as well as the Distance Ratio Divergence of both best-scoring colormaps are among the best, emphasizing a good perceptual linearity. The Attention Steering score of *Bremm et al. (regular)* is among the best rankings and thus outperforms *TeulingFig3NoWhitening*. In summary, the static 2D colormaps *Bremm et al. (regular)* and *TeulingFig3NoWhitening* score best in the use case section. Both colormaps produce excellent rankings for the relevant tasks **SR1** or **SR2** shown in Table 3. At the same time the colormaps do not show any weaknesses in any of the underlying quality assessment measures (cf. Table 2). In this regard both colormaps differ from other well-ranked colormaps which, however, suffer from deficiencies in single relevant quality assessment measures and thus were excluded.



Figure 4. An appropriate colormap for the use case: Bremm et al. (regular). A sufficient number of JND Colors, distances to white and black, as well as perceptual linearity scores above average, and a good Attention Steering score.

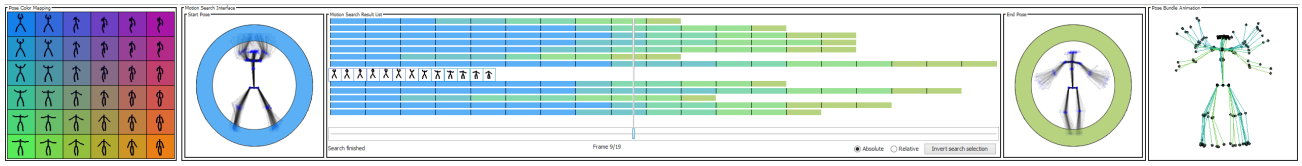


Figure 5. An appropriate colormap for the use case: Teuling3NoWhitening. More JND Colors than Bremm et al. (regular), distances to white and black, as well as perceptual linearity scores above average.

6. DISCUSSION AND FUTURE WORK

Reflections on the Test Results The three elementary task-requirements (**ER1**, **ER2**, and **ER3**) each emphasize completely different design goals for 2D colormaps, which confirms the challenges of design trade-offs described in literature. For the synoptic tasks (**SR1**, **SR2**, **SR3**, and **SR4**), however, the results are more homogeneous. Here, 2D colormaps which mastered these trade-offs best prevailed. Altogether, ten colormaps have proven to be very robust, passing all described analysis tasks without any exclusion for either a black or a white background. Two of these colormaps are based on L*a*b while the other colormaps are designed in sRGB/HSV. The title Figure 1 shows the eight 2D colormaps having a positive sum of score points with the JND regions drawn ($\Delta E = 5$). Assessing these local quality aspects of 2D colormaps in detail may be a subject of future work.

Discussion In this paper, we analyzed the applicability of static 2D colormaps with respect to a set of tasks which can be tackled by mapping multivariate data to color. We neglected the class of colormaps which were designed explicitly for mapping bivariate data (see Figure 2). A point of discussion is that for some presented colormaps, insufficient information for a re-implementation was provided. In these cases, we applied work-around techniques to implement the colormaps. One strategy was making use of regression models based on available color distributions (used for Robertson,[57] Lespintas,[39] Griscom,[43] and Baum et al.[20,36]). Despite the fact that we did not detect considerable differences between our implementations and the 2D colormaps printed in the original papers visually, we would prefer to re-implement the 2D colormaps on the basis of the

original design specifications. A similar issue relates to the sRGB-based test environment where the measures were taken. We know that in some cases, conversions to sRGB lead to lossy conversions which impacts the applicability to non-screen uses and distorts the value of the quality measures as such. However, although other implementers may use any kind of strategy to deal with these lossy conversions, our simple clipping algorithm may well be the first choice of many. In future work, this issue may be addressed more rigorously. Looking back to the colormap design taxonomy of Tominski et al.,[12] we made assumptions in order to reduce the number of dependent variables. We determined the *where* dimension (displays, printers, or other devices) as static with the focus on high resolution displays and print media with the assumption of a white or a black background color. Similar to the latter, the choice of accent colors e.g., for text, contours, or borders may influence the choice of a 2D colormap. In cases where white and black colors are reserved for contours and background (like in the MotionExplorer system) both the white distance and the black distance measure can be utilized. Colormaps performing well for both black and white distances are *Bremm et al. (regular)*, *Mittelstädt et al.*, *TeulingFig3NoWhitening*, and *TeulingFig4a*. Highly related to the latter concern is the use of the alpha-value of colors as an additional visual variable which is typically utilized to indicate data-specific variances, or uncertainty in general. The black distance, the white distance, and the attention steering measure may serve as starting points for a more in-depth discussion about quality measures adressing the use of the alpha compositing. Finally, we want to emphasize the *what* dimension in the taxonomy of Tominski et al.[12] Another class of 2D colormaps exists which explicitly incorporates intrinsic data properties for the design of 2D colormaps.[11,64] These 2D colormaps are data-adaptive variants and only exist in combination with a particular data set, in contrary to the static 2D colormaps tested for this paper.

Future Work We already listed a couple of concerns which may be subject of future work. We also want to focus on a full-automatic ranking of colormaps with respect to a given analysis task. Here, developing functions to weight the provided quality assessment measures on the basis of linear combinations is conceivable. Similarly to other retrieval algorithms, the question arises whether the results of the single quality assessment measures should be utilized on value-based, or on rank-based criteria. A related subject of future work concerns the development of task-specific measures, e.g., by deriving measures based on task-specific sampling strategies. Another possible extension would be the visual-interactive analysis of given 2D colormaps for global, but also for local quality assessment measures. Closely related is the development of decision support capability for the choice of 2D colormaps. It may also be a possible concern not only to evaluate existing, but also to *create new* or *adapt* 2D colormaps based on particular data, tasks, and usage parameters. Finally, we aim to focus on data-adaptive 2D colormaps and compare this special class of 2D colormaps with the static variants surveyed in this work. It will be interesting to see advantages and disadvantages of different 2D colormap classes in detail.

7. CONCLUSION

In this paper, we presented a survey of static 2D colormaps with the overall goal to assess the quality of existing variants with respect to well-known analysis tasks. To this end, we re-implemented the most prominent designs and variants, covering the 2D colormaps of over 50 related works in total. The provided colormaps are based on the sRGB color space, variants of CIE L*a*b* and HSV. We provided a set of different quality assessment measures which we mapped on seven different analysis tasks for color-coded multivariate data. A test environment enabled us to present detailed quality information for every 2D colormap and every quality assessment measure. Finally, we made suggestions for the choice of 2D colormaps for each of the seven tasks. We hope that the information visualization community, but also affine application domains may benefit from the quality assessment results, the task-based guidelines, and the discussion presented in this work.

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